

TO RECORD SIMPLE MUSCLE TWICH (SMT)

**MUSCLE
TWITCH**



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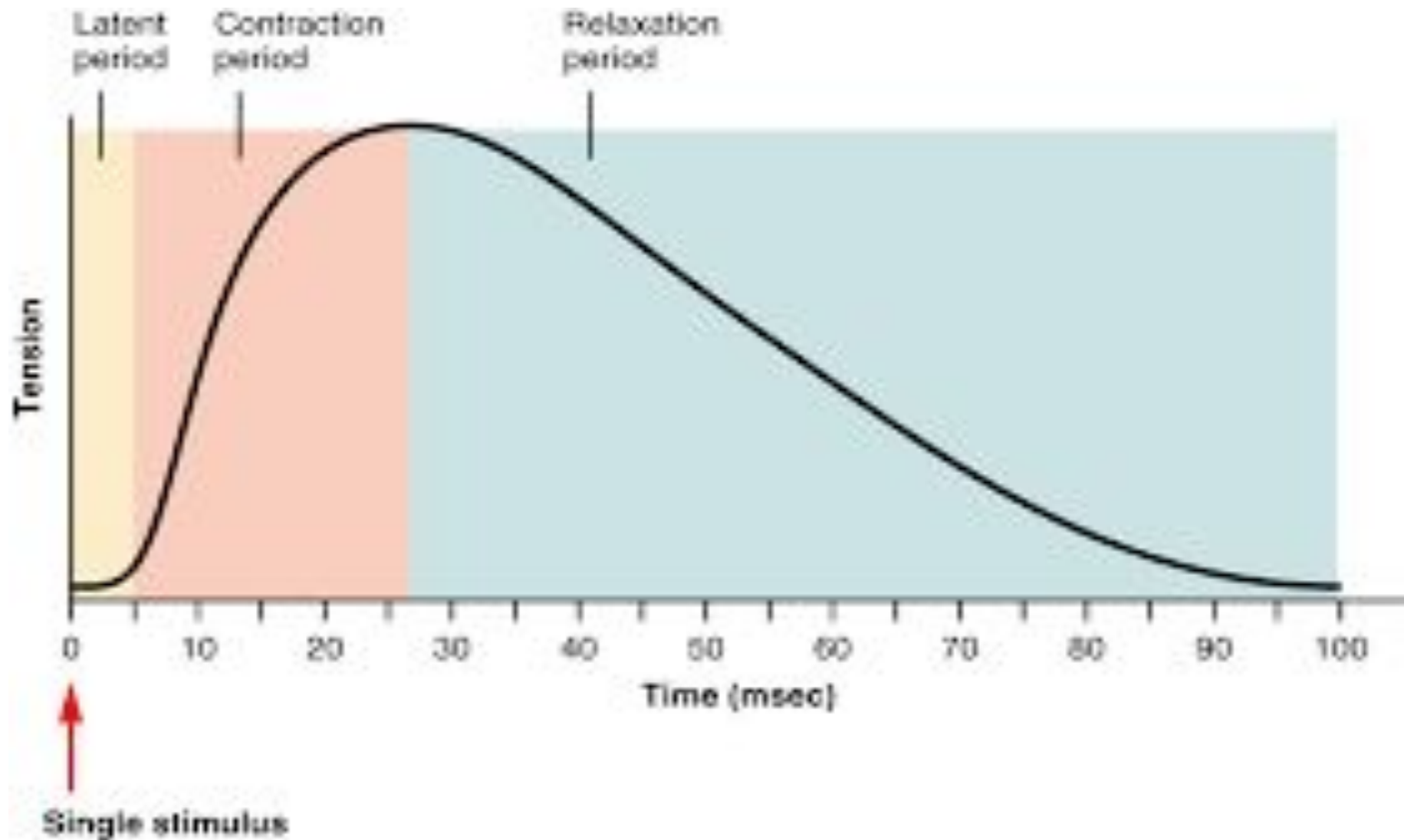
Simple muscle twitch

The simple muscle twitch (S.M.T) may be define as the response of the muscle to a single stimulus applied to a muscle, it give rise to a sudden contraction for a fraction of time.

S.M.T consist of three phases

- **1. LATENT PERIOD**
- **2. CONTRACTION PERIOD**
- **3. RELEXATION PERIOD**

Graph presentation of SMT



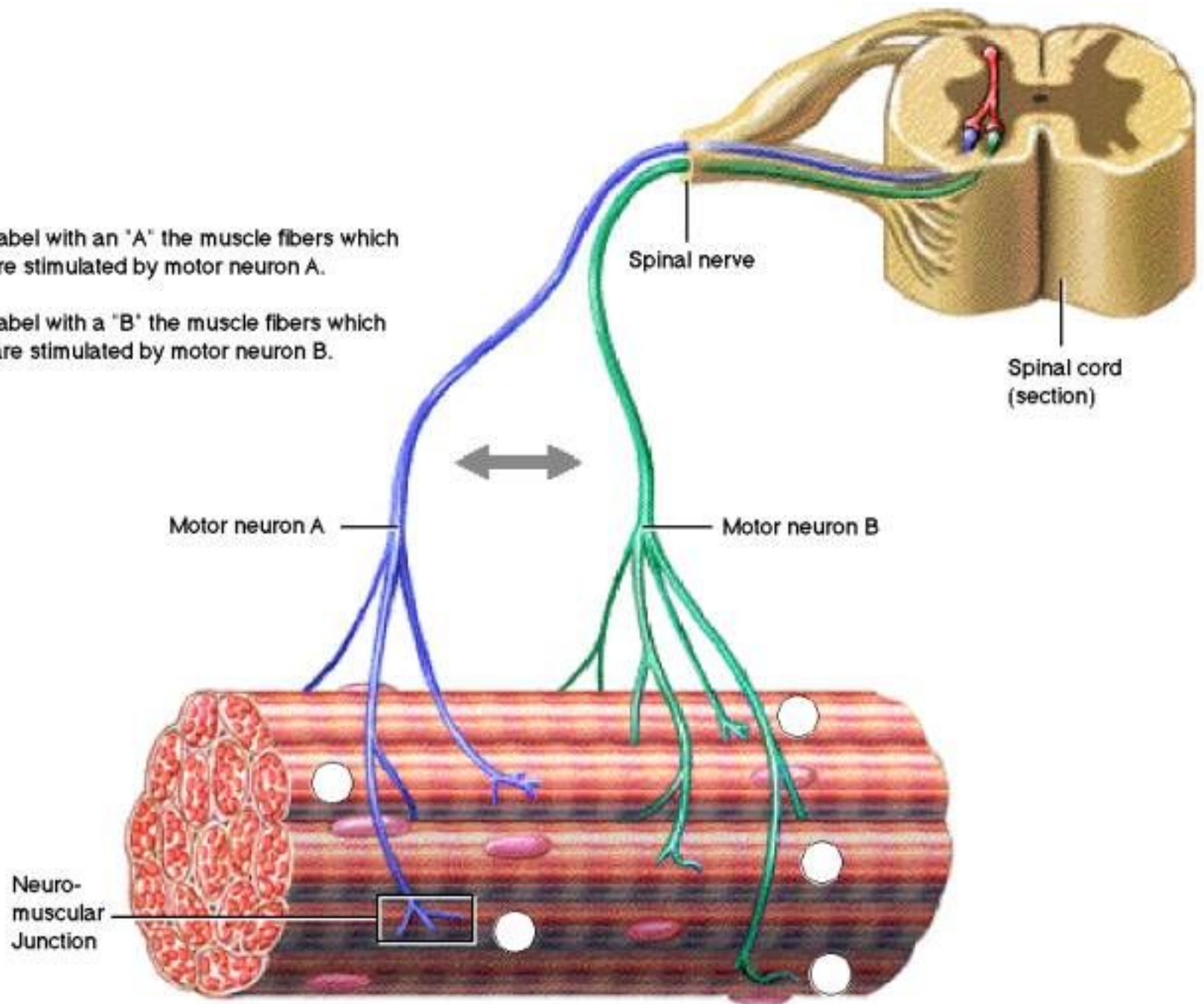
1-Latent Period

A phase following the point of stimulation i.e. from point of stimulation (PS) to point of contraction (PC) during which no apparent changes takes place in the muscle.

- During this period the action potential is generated in the nerve, this impulse propagates towards the neuromuscular junction, transmitted from the nerve to muscle fibers and prepare themselves for the contraction.
- The duration of L.P is about 0.01 second.

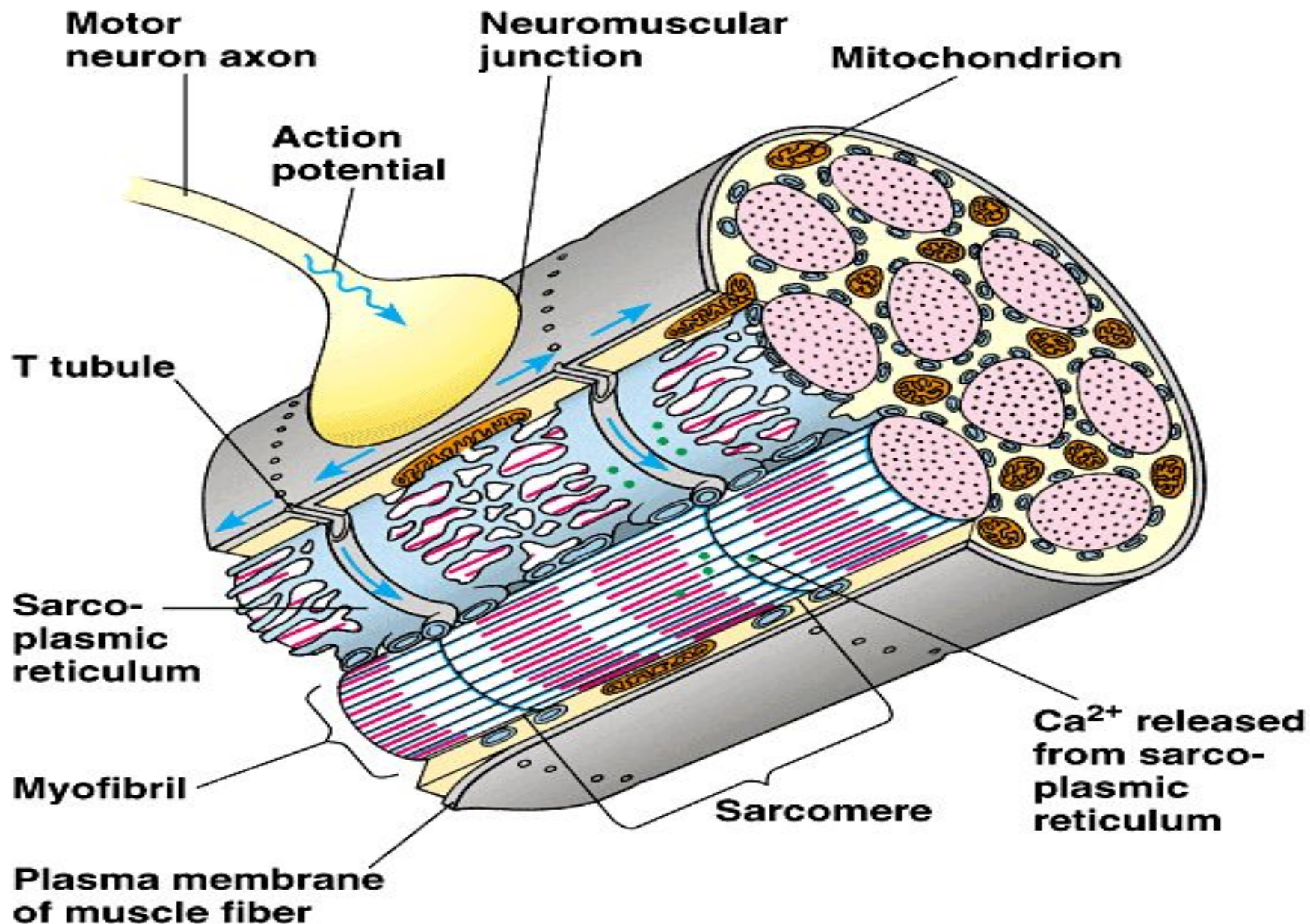
Label with an "A" the muscle fibers which are stimulated by motor neuron A.

Label with a "B" the muscle fibers which are stimulated by motor neuron B.



2.CONTRACTION PERIOD

A phase of shortening and contraction of muscle during which the actin and myosin filaments slides on each other and the muscle shortens in length. The duration of this period is about 0.04 seconds.



3. RELEXATION PERIOD

A phase of returning or relaxation during which the muscle returns to its original length. Its duration is about 0.05 seconds.

- **Threshold**
- **Sub threshold**
- **Suprathreshold**

- **Threshold ? ?**

The threshold stimulus is the stimulus required to create an action potential.

The minimal strength of stimulation needed to elicit a **muscle contraction** is called the **threshold**

So any stimulus under this level will not cause muscle contraction, while a stimulus above this level will cause the muscle to contract.

The higher the stimulus the more muscle fibers are recruited, and thus the higher the response.

- **Sub threshold stimulus ? ? ?**

- A stimulus too weak to evoke a response.

- **Suprathreshold ? ?**

- Stimuli above a specified threshold

REQUIREMENTS:-

- Nerve muscle preparation
- Lab tutor, Stand, Ringer's solution or normal saline,
- Pins
- dissecting instruments.

PROCEDURE

- Make sure the muscle is moist and is in contact with the stimulating electrodes.
- In the Stimulator panel check that the voltage is set to 50 mV.
- Click Start. The software is set up so that Lab Tutor will now record for 625 ms with a single stimulus being delivered after an initial delay of 50 ms.

- The contractile response of the muscle and the time at which the stimulus was delivered are shown in the Lab Tutor panel.
- Lab Tutor will also enter a comment showing the amplitude of stimulation.
- Continue to increase the stimulus amplitude in 50 mV steps, recording the responses until you have 3 successive stimuli that do not produce any further increase in the amplitude of the muscle contraction

RESULTS

S.NO.	PHASES	STANDARD VALUES	OBSERVE VALUES
1.	Latent Period (L.P)	0.01 sec.	
2.	Contraction Period (C.P)	0.04 sec.	
3.	Relaxation Period (R.P)	0.05 sec.	
4.	Height of Contraction		

PRECAUTIONS

- ❑ Nerve and muscle preparation should remain wet with ringer's solution.
- ❑ Nerve and muscle preparation should be place properly on stand.
- ❑ Lab tutor is properly assemble.
- ❑ Electrode is placed over muscle fibers.

Relevant Questions

Q-1-Define simple muscle twitch.

- It is the response of the muscle to a single maximal stimulus

Q-2-What is excitation-contraction coupling?

It is Excitation–contraction coupling is the physiological process of converting an electrical stimulus to a mechanical response. It is the link (transduction) between the action potential generated in the sarcolemma and the start of a muscle contraction

Q-3- During latent period, what is happening?

During the latent period of a muscle twitch, the action potential propagates along the sarcolemma, calcium ions are released from the sarcoplasmic reticulum, and excitation-contraction coupling occurs, but visible muscle contraction hasn't yet begun.

Q-4-What is the normal duration of latent period, contraction phase and relaxation phase?

The duration of L.P is about 0.01 second.

The duration of contraction period is about 0.04 seconds.

The duration of Relaxation period is 0.05 sec

Q-5-What is the resting membrane potential of skeletal muscle?

- The resting membrane potential of skeletal muscle is typically around -90 mV,

Q-6- What is the threshold potential of skeletal muscle?

The threshold potential of skeletal muscle, the point at which an action potential is triggered, is typically around -50 to -55 mV.